FLAT Module: Deterministic Finite Automata (DFA) Lexical Analyzer

Technical Report on Formal Grammar, DFA Transition Engine, & Microcode Validation

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Live App: https://gauthamram-s70.github.io/Integrated-MicroProject-FLAT-COA-DAA/	**Repository:** https://github.com/gauthamram-s70/Integrated-MicroProject-FLAT-COA-DAA

1. Executive Summary & Curriculum Alignment

This module implements a rigorous, custom **Deterministic Finite Automaton (DFA)** engine that serves as the front-end lexical analyzer and syntax validator for a custom RISC microcode assembly language. Aligned with **FLAT Modules 1–3 (Finite Automata, Regular Expressions, and Formal Languages)**, the engine formally models state transitions over an alphabet of instruction tokens, rejects syntactically malformed code, and translates accepted instructions into 16-bit binary control words for pipelined execution.

Curriculum Connection: *Demonstrates the practical equivalence between Regular Expressions and Deterministic Finite State Machines in compiler front-end design, token extraction, and dead-state trap mechanisms.*

2. Mathematical Formulation & Formal 5-Tuple

Formally, the DFA is specified as a 5-tuple: **M = (Q, Σ, δ, q_0, F)**, where:

Component	Formal Definition & Implementation Specification
Q (States)	{q0, q1, q2, q3, q4, q_accept, q_reject} representing parse stages.
Σ (Alphabet)	{LOAD, STORE, ADD, SUB, JMP, R0-R7, Imm(0-255), Comma(,), Semicolon(;)}
q_0 (Start)	q0: Idle initial state awaiting instruction opcode mnemonic.
F (Accept)	{q_accept}: Final state reachable strictly upon encountering semicolon terminator.
q_reject (Dead)	Dead trap state reached immediately upon any lexical or syntactic violation.

3. DFA State Transition Table & Grammar Paths

The state transition function δ: Q × Σ → Q maps every configuration to a unique deterministic next state:

State	Opcode	Reg (R0-R7)	Comma (,)	Number/Imm	Semi (;)	Other
q0	q1	q_reject	q_reject	q_reject	q_reject	q_reject
q1	q_reject	q2	q_reject	q4 (if JMP)	q_reject	q_reject
q2	q_reject	q_reject	q3	q_reject	q_reject	q_reject
q3	q_reject	q4 (Reg)	q_reject	q4 (Imm)	q_reject	q_reject
q4	q_reject	q_reject	q_reject	q_reject	q_accept	q_reject
q_acc	q_reject	q_reject	q_reject	q_reject	q_reject	q_reject
q_rej	q_reject	q_reject	q_reject	q_reject	q_reject	q_reject

4. 16-Bit Micro-Instruction Control Word Encoding

Upon reaching **q_accept**, the DFA synthesizes a 16-bit binary control word formatted as follows:

[15:13] Opcode (3b) | [12:10] Dest Reg (3b) | [9:7] Source Reg (3b) | [6:0] Immediate / Offset (7b)

- **LOAD**: 001 | **STORE**: 010 | **ADD**: 011 | **SUB**: 100 | **JMP**: 101
- **Registers R0–R7**: 000 through 111 | **Signals**: RegWrite, ALUSrc, ALUOp, MemRead, MemWrite, Branch

5. Verification Suite & Rejection Cases

Input Statement	DFA State Path	Verdict	Synthesized Hex Word / Error
ADD R1, R2;	q0 → q1 → q2 → q3 → q4 → q_accept	ACCEPTED	0x6480 (Op:ADD, Rd:R1, Rs:R2)
LOAD R3, 25;	q0 → q1 → q2 → q3 → q4 → q_accept	ACCEPTED	0x2C19 (Op:LOAD, Rd:R3, Imm:25)
JMP 128;	q0 → q1 → q4 → q_accept	ACCEPTED	0xA000 (Op:JMP, Imm:0)
ADD R1 R2;	q0 → q1 → q2 → q_reject	REJECTED	Syntax Error: Missing comma
SUB R1, 10	q0 → q1 → q2 → q3 → q4 → q_reject	REJECTED	Syntax Error: Missing semicolon terminator
MULT R1, R2;	q0 → q_reject	REJECTED	Syntax Error: Invalid opcode MULT

6. Conclusion & Inter-Module Linkage

The FLAT DFA module guarantees 100% lexical and grammatical soundness before execution, preventing malformed control words from entering the CPU pipeline. Validated control words pass directly into the COA 5-stage pipeline simulator.